Bias Analysis Report

Executive Summary

This report provides an overview of bias in the dataset, highlighting key findings, potential risks, and recommended actions. Each section presents an analysis of bias, visualizations, and actionable insights.

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1. Data Profiling Summary

This section provides an overview of the dataset, including the types and number of variables, potential missing values, and initial statistical assessments.

Data Profiling Alerts

Total Variables: 21

Numeric Features: 21

Categorical Features: 0

- [CRC_incidence_val_Percent] 201 (8.2%) missing values

- [Alcohol use_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet high in processed meat_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet high in red meat_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet high in sodium_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet high in sugar-sweetened beverages_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet high in trans fatty acids_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in calcium_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in fiber_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in fruits_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in legumes_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in milk_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in nuts and seeds_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in polyunsaturated fatty acids_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in vegetables_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Diet low in whole grains_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [High body-mass index_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Low physical activity_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Smoking_Rate_Summary exposure value_val] 411 (16.8%) missing values

- [Unsafe water source_Rate_Summary exposure value_val] 411 (16.8%) missing values

Kurtosis bar plot

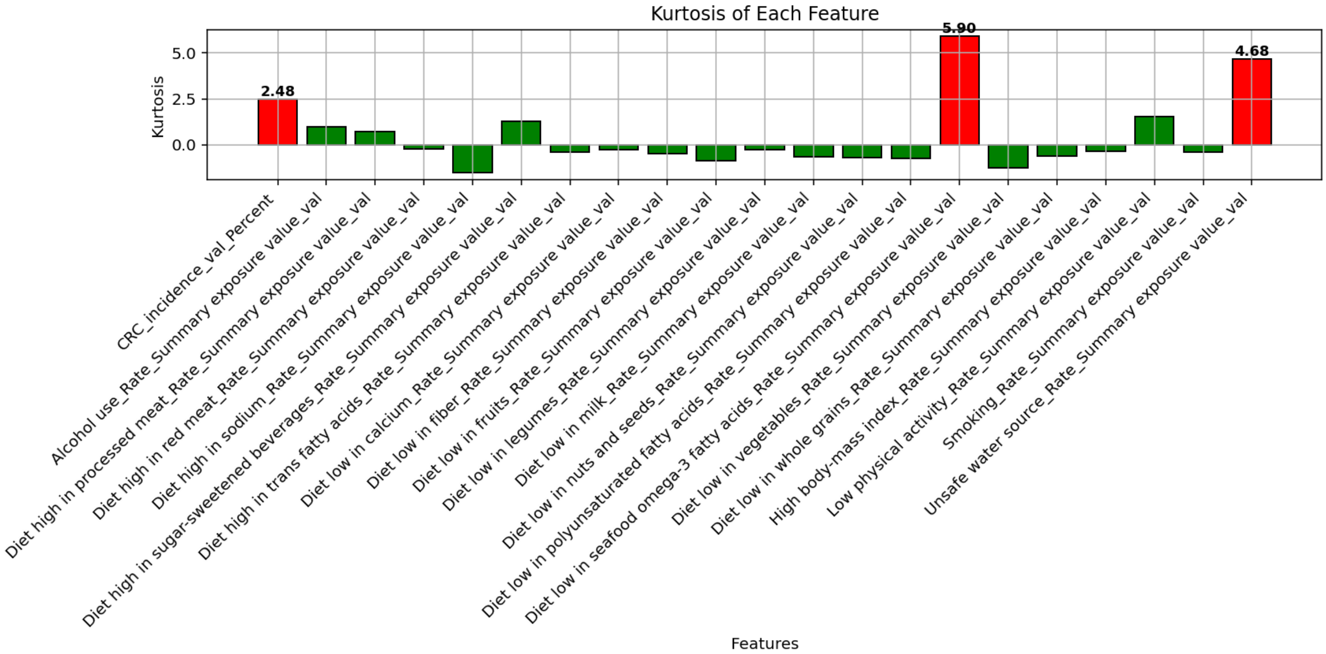

Figure: Kurtosis bar plot. This plot shows the kurtosis values for each feature, highlighting the statistical distribution characteristics.

Skewness bar plot

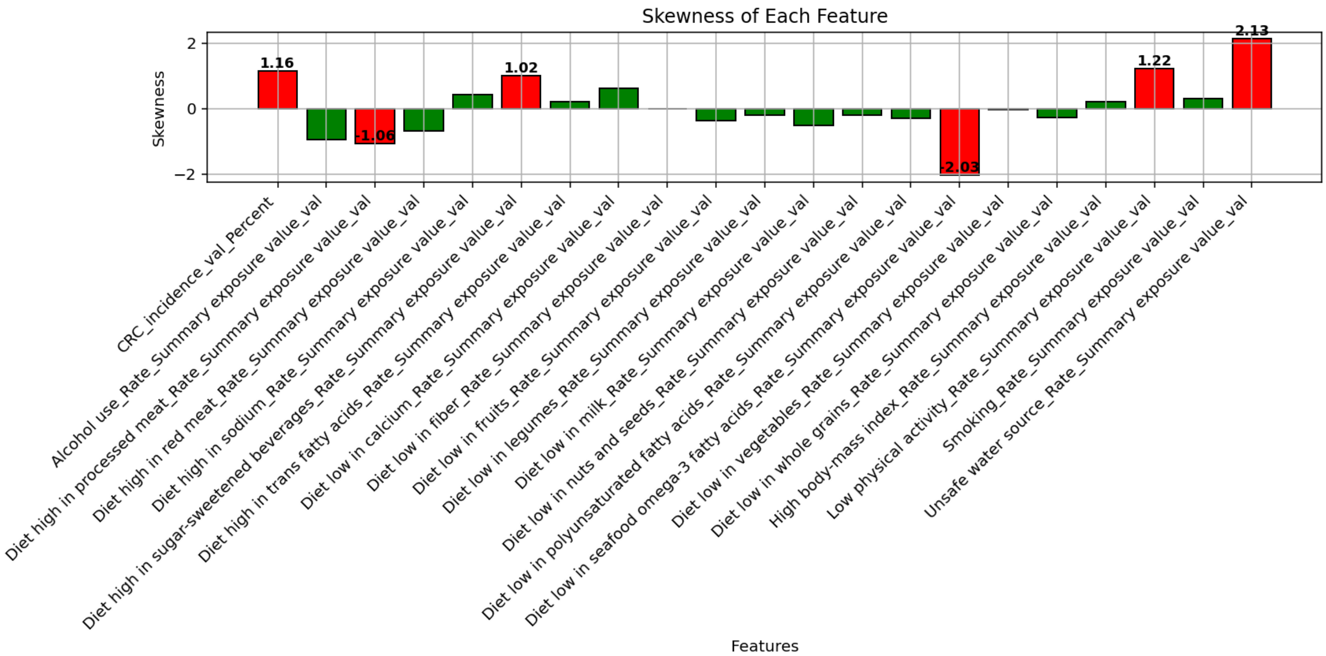

Figure: Skewness bar plot. This plot shows the skewness values for each feature, highlighting the statistical distribution characteristics.

2. Bias Alerts and Key Findings

This section highlights specific bias risks identified in the dataset, categorized by severity levels.

- Severe bias detected in 'CRC_incidence_val_Percent'. Consider re-evaluating data collection methods.

- Moderate bias detected in 'Alcohol use_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Severe bias detected in 'Diet high in processed meat_Rate_Summary exposure value_val'. Consider re-evaluating data collection methods.

- Moderate bias detected in 'Diet high in red meat_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Severe bias detected in 'Diet high in sodium_Rate_Summary exposure value_val'. Consider re-evaluating data collection methods.

- Severe bias detected in 'Diet high in sugar-sweetened beverages_Rate_Summary exposure value_val'. Consider re-evaluating data collection methods.

- Moderate bias detected in 'Diet high in trans fatty acids_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Moderate bias detected in 'Diet low in calcium_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Mild bias detected in 'Diet low in fiber_Rate_Summary exposure value_val'. Monitor for localized bias but no immediate action needed.

- Mild bias detected in 'Diet low in fruits_Rate_Summary exposure value_val'. Monitor for localized bias but no immediate action needed.

- Moderate bias detected in 'Diet low in legumes_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Mild bias detected in 'Diet low in milk_Rate_Summary exposure value_val'. Monitor for localized bias but no immediate action needed.

- Moderate bias detected in 'Diet low in nuts and seeds_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Mild bias detected in 'Diet low in polyunsaturated fatty acids_Rate_Summary exposure value_val'. Monitor for localized bias but no immediate action needed.

- Severe bias detected in 'Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val'. Consider re-evaluating data collection methods.

- Moderate bias detected in 'Diet low in vegetables_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Severe bias detected in 'Diet low in whole grains_Rate_Summary exposure value_val'. Consider re-evaluating data collection methods.

- Moderate bias detected in 'High body-mass index_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Severe bias detected in 'Low physical activity_Rate_Summary exposure value_val'. Consider re-evaluating data collection methods.

- Moderate bias detected in 'Smoking_Rate_Summary exposure value_val'. Investigate potential data skew or imbalance.

- Severe bias detected in 'Unsafe water source_Rate_Summary exposure value_val'. Consider re-evaluating data collection methods.

Figures below illustrate the divergence metrics used to assess data consistency.

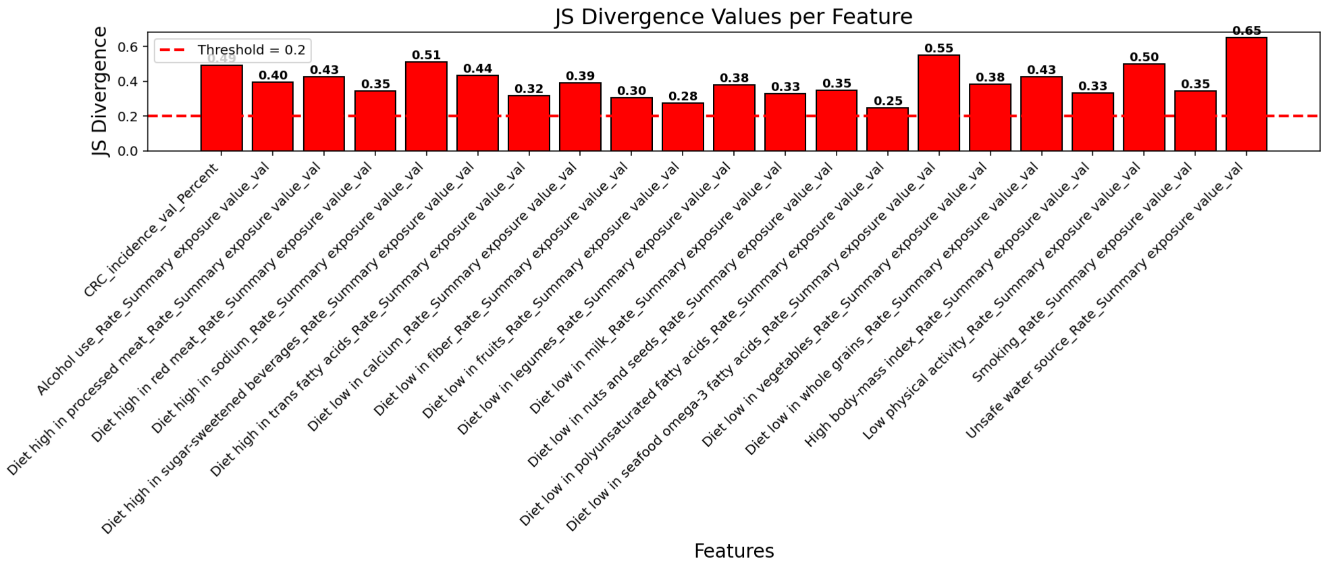

Figure: Js divergence bar plot

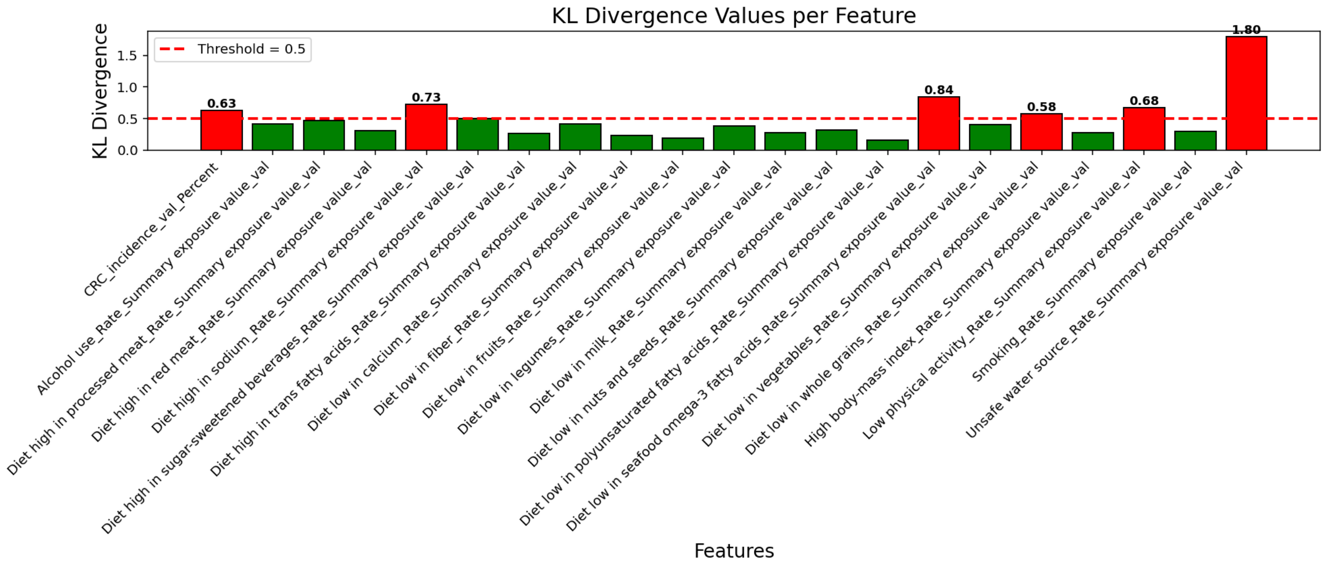

Figure: Kl divergence bar plot

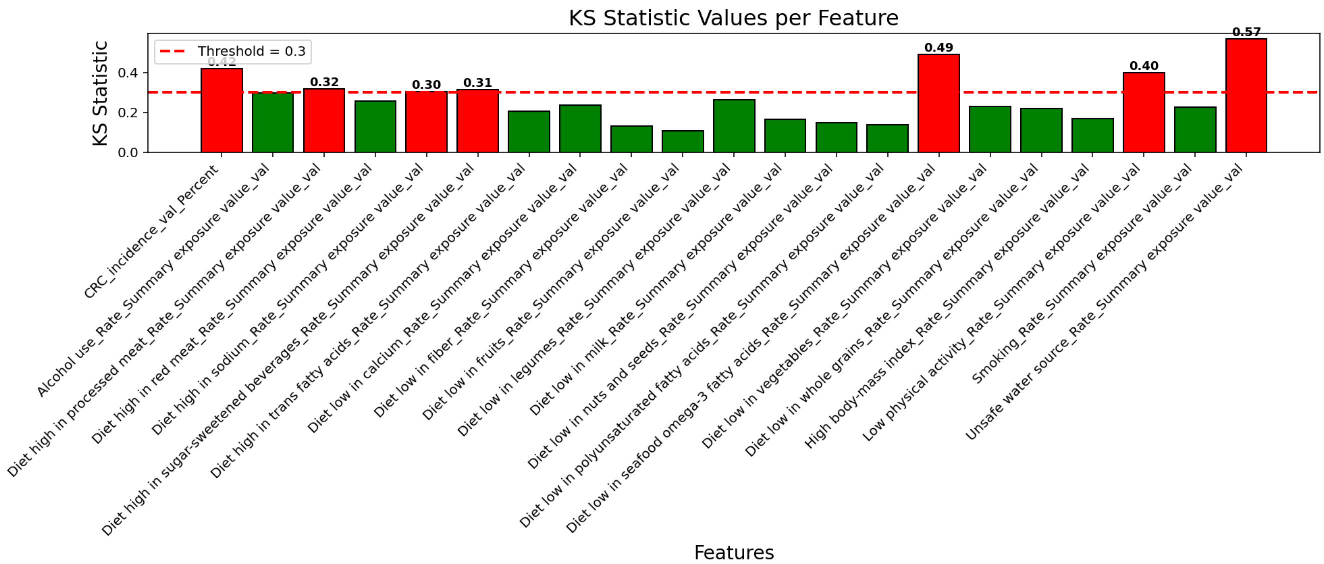

Figure: Ks statistic bar plot

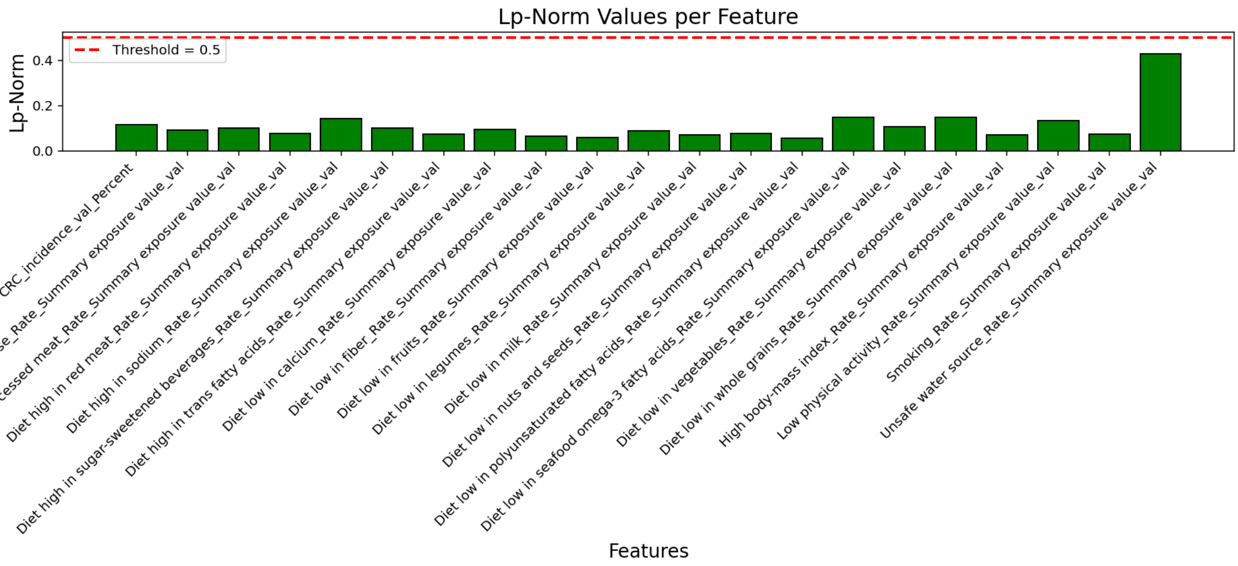

Figure: Lpnorm bar plot

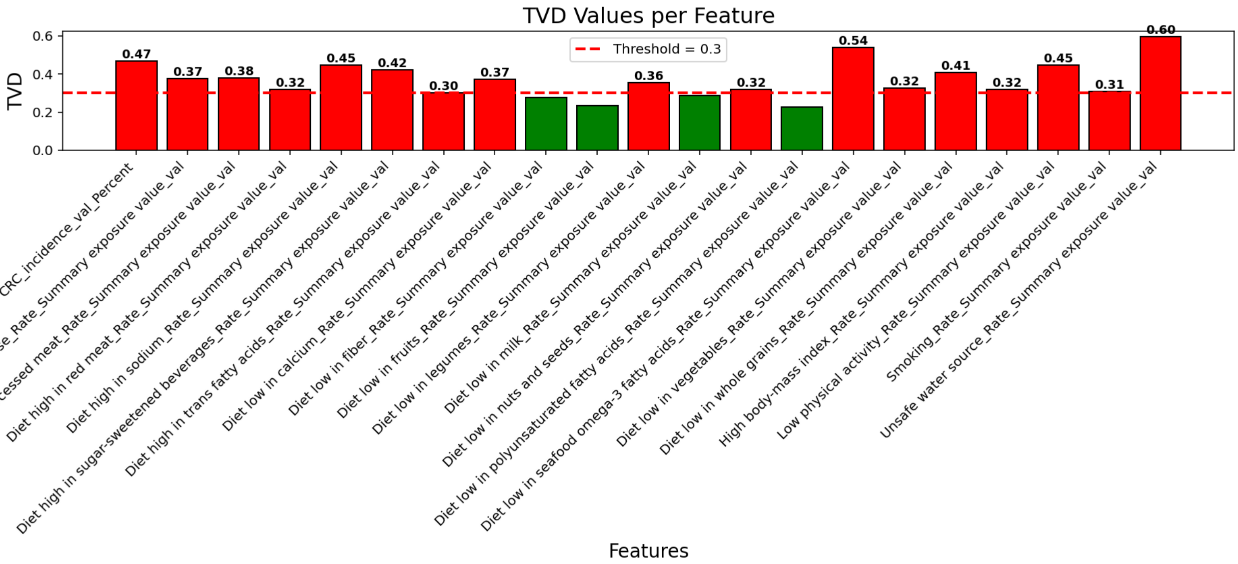

Figure: Tvd bar plot

SUMMARY - Features Flagged by 3+ Metrics

- CRC_incidence_val_Percent (Flagged by 4 metrics)
- Diet high in sodium_Rate_Summary exposure value_val (Flagged by 4 metrics)
- Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val (Flagged by 4 metrics)
- Low physical activity_Rate_Summary exposure value_val (Flagged by 4 metrics)
- Unsafe water source_Rate_Summary exposure value_val (Flagged by 4 metrics)
- Diet high in processed meat_Rate_Summary exposure value_val (Flagged by 3 metrics)
- Diet high in sugar-sweetened beverages_Rate_Summary exposure value_val (Flagged by 3 metrics)
- Diet low in whole grains_Rate_Summary exposure value_val (Flagged by 3 metrics)

Top 5 Most Biased Features Per Metric

KL Divergence

- Unsafe water source_Rate_Summary exposure value_val (KL Divergence: 1.796)

- Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val (KL Divergence: 0.842)

- Diet high in sodium_Rate_Summary exposure value_val (KL Divergence: 0.727)

- Low physical activity_Rate_Summary exposure value_val (KL Divergence: 0.676)

- CRC_incidence_val_Percent (KL Divergence: 0.630)

JS Divergence

- Unsafe water source_Rate_Summary exposure value_val (JS Divergence: 0.651)

- Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val (JS Divergence: 0.551)

- Diet high in sodium_Rate_Summary exposure value_val (JS Divergence: 0.510)

- Low physical activity_Rate_Summary exposure value_val (JS Divergence: 0.498)

- CRC_incidence_val_Percent (JS Divergence: 0.491)

KS Statistic

- Unsafe water source_Rate_Summary exposure value_val (KS Statistic: 0.570)

- Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val (KS Statistic: 0.491)

- CRC_incidence_val_Percent (KS Statistic: 0.421)

- Low physical activity_Rate_Summary exposure value_val (KS Statistic: 0.399)

- Diet high in processed meat_Rate_Summary exposure value_val (KS Statistic: 0.320)

TVD

- Unsafe water source_Rate_Summary exposure value_val (TVD: 0.596)

- Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val (TVD: 0.539)

- CRC_incidence_val_Percent (TVD: 0.470)

- Diet high in sodium_Rate_Summary exposure value_val (TVD: 0.446)

- Low physical activity_Rate_Summary exposure value_val (TVD: 0.446)

Lp-Norm

- Unsafe water source_Rate_Summary exposure value_val (Lp-Norm: 0.429)

- Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val (Lp-Norm: 0.147)

- Diet low in whole grains_Rate_Summary exposure value_val (Lp-Norm: 0.147)

- Diet high in sodium_Rate_Summary exposure value_val (Lp-Norm: 0.142)

- Low physical activity_Rate_Summary exposure value_val (Lp-Norm: 0.133)

High-Dimensional Bias Analysis

High-Dimensional Bias Analysis examines bias patterns that may not be evident in lower dimensions. Techniques such as community detection, anomaly detection, and unsupervised learning methods are employed to reveal hidden structures in the data. This section covers key insights obtained from these methods.

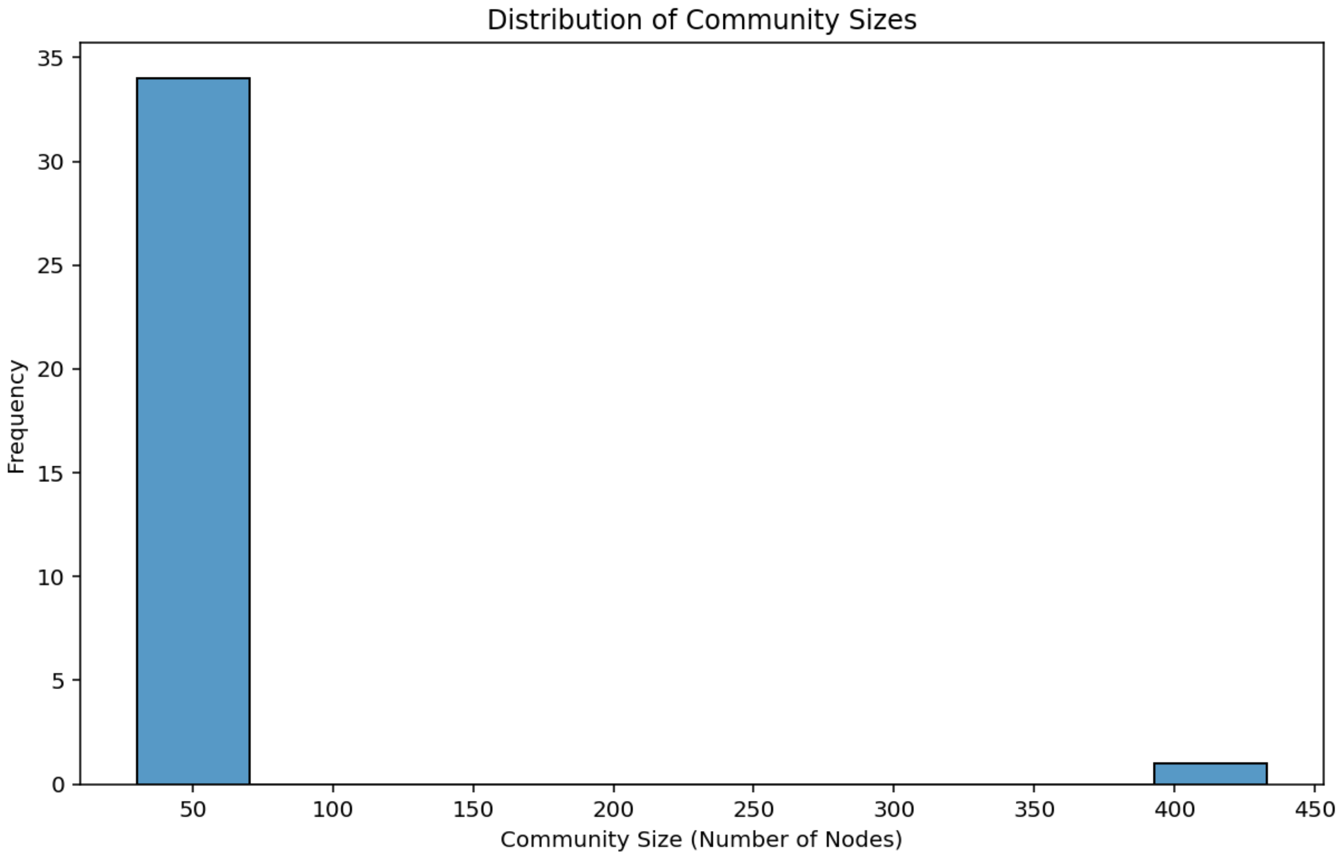

Figure: Community Size Distribution: This histogram illustrates the frequency of detected community sizes in the dataset. A highly imbalanced distribution suggests that some communities are disproportionately large or small, which could indicate structural bias in the network representation of the data.

Community Detection from Similarity Graph

Figure: Community Detection Graph: This visualization represents the identified communities in the similarity network. Different colors correspond to distinct communities. If certain groups are overly connected or isolated, this may signal potential biases in how data points relate to one another.

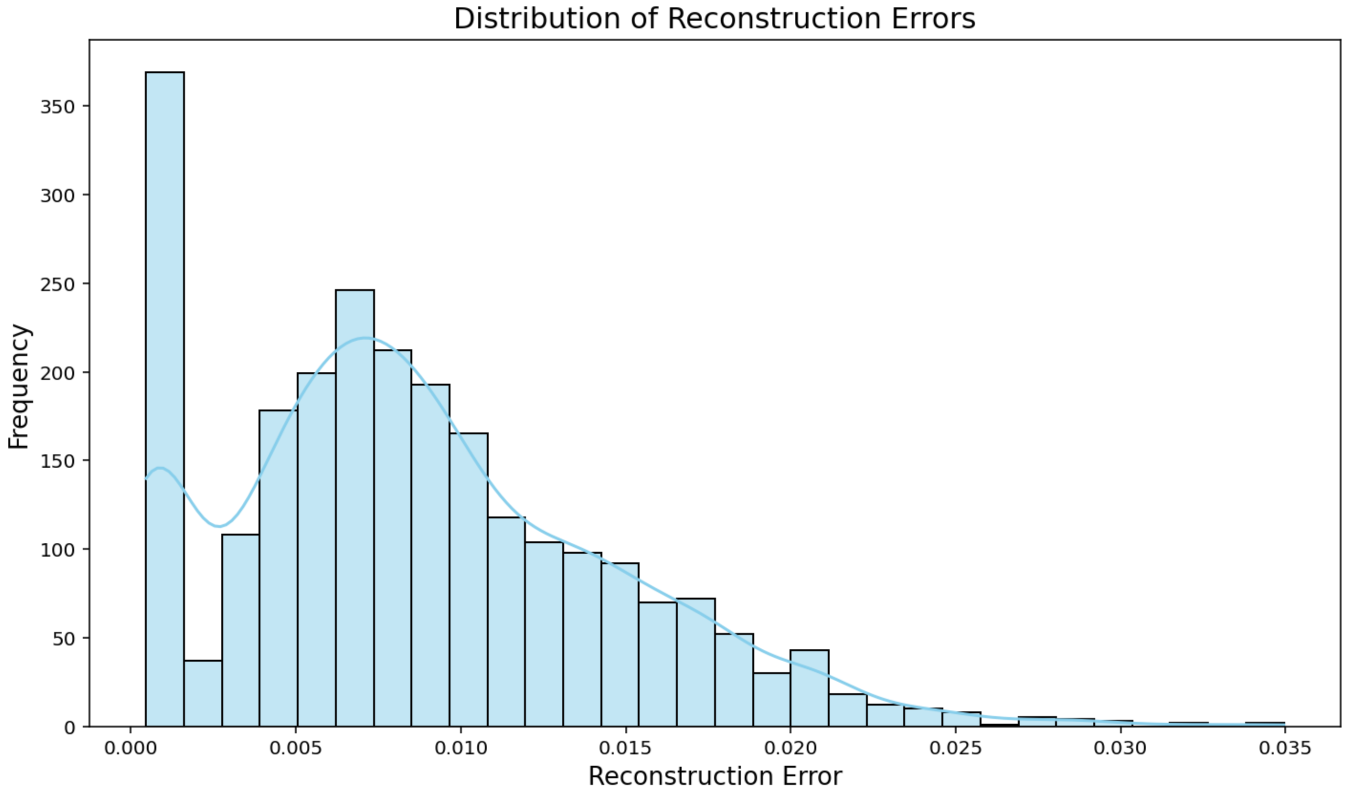

Figure: Reconstruction Error Distribution (Autoencoder): This plot shows the distribution of reconstruction errors. Higher errors indicate data points that do not conform to the learned patterns, suggesting possible anomalies. If errors are concentrated in specific demographic groups, this may reveal bias.

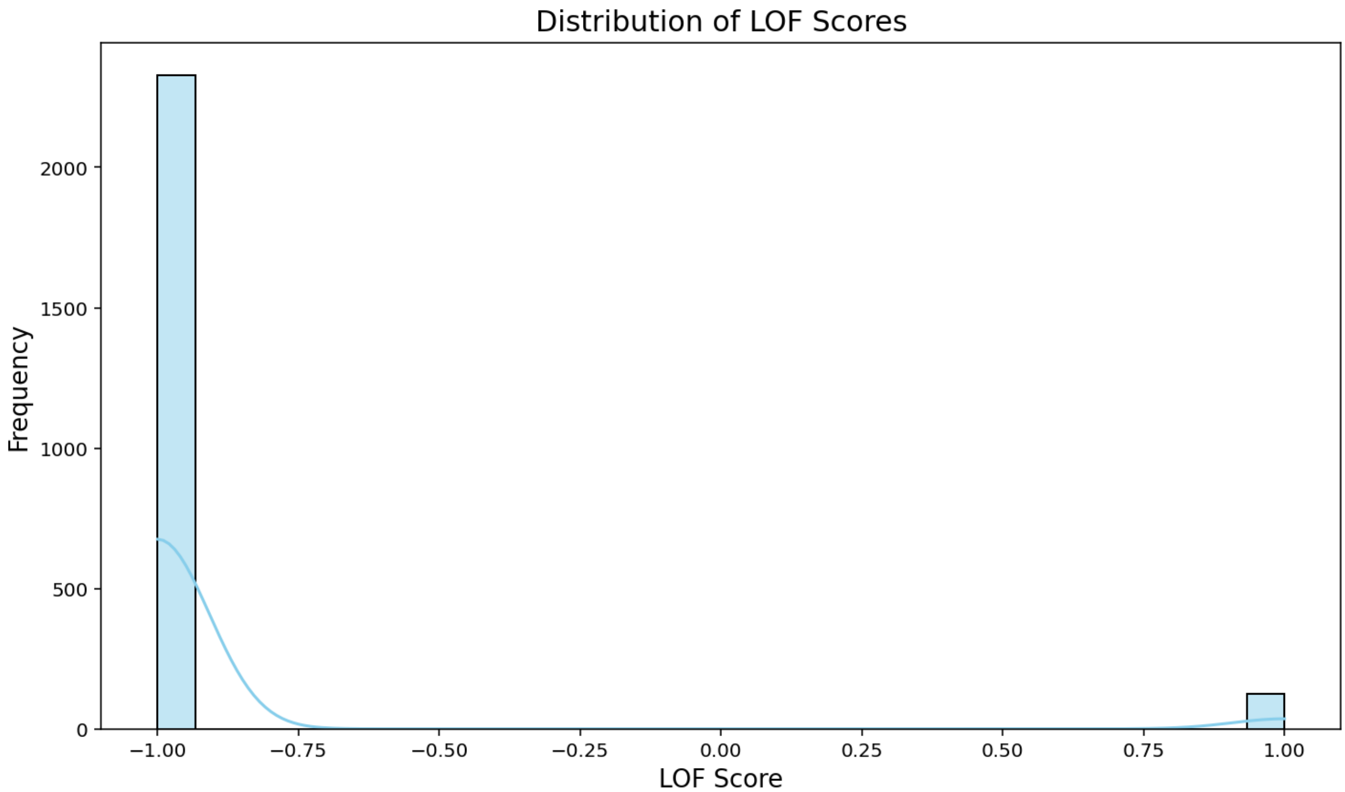

Figure: LOF Score Distribution: Local Outlier Factor (LOF) identifies data points that differ significantly from their neighbors. A skewed LOF distribution may indicate that some features or groups exhibit distinct patterns, suggesting potential bias.

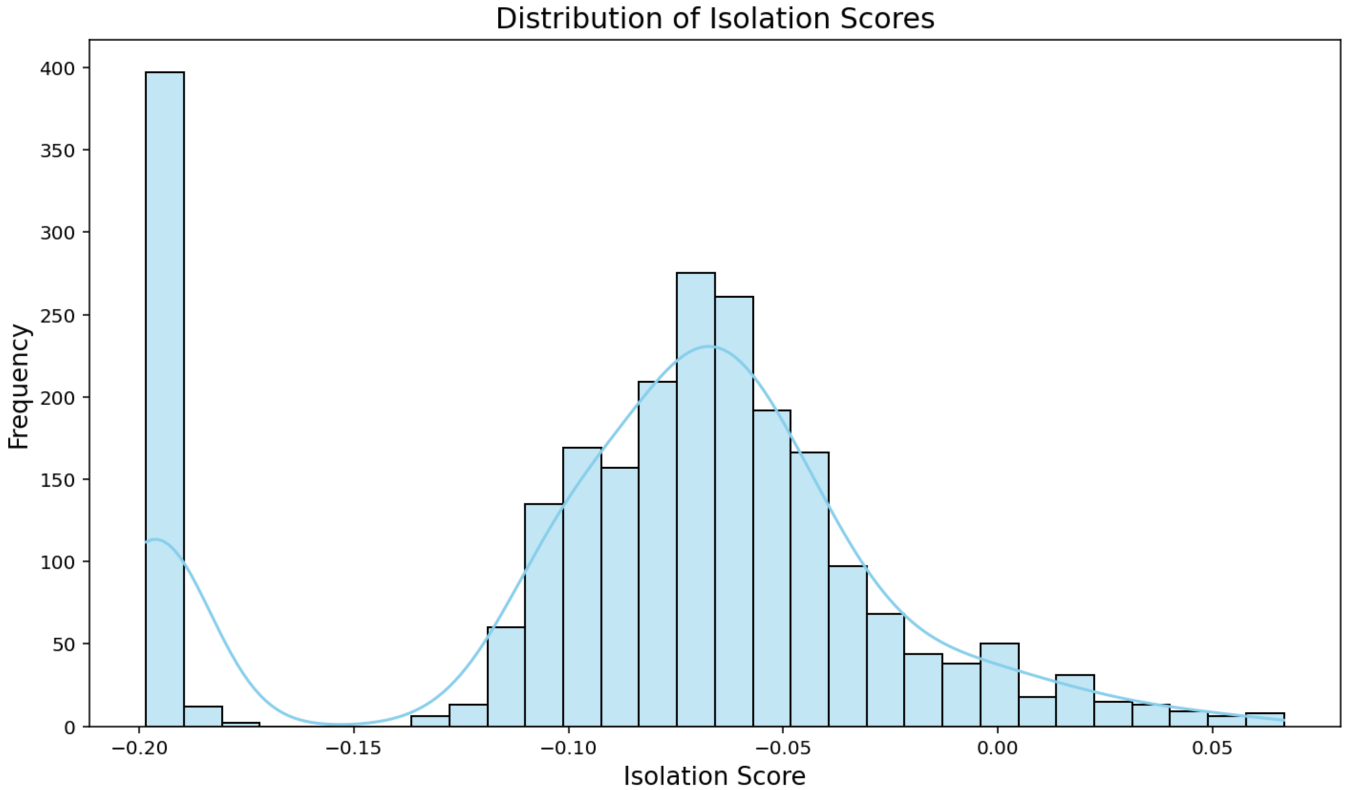

Figure: Isolation Score Distribution: This histogram displays the anomaly scores from Isolation Forest, a method for detecting outliers. High isolation scores indicate features that are easily separable, which may suggest bias-driven segmentation in the dataset.

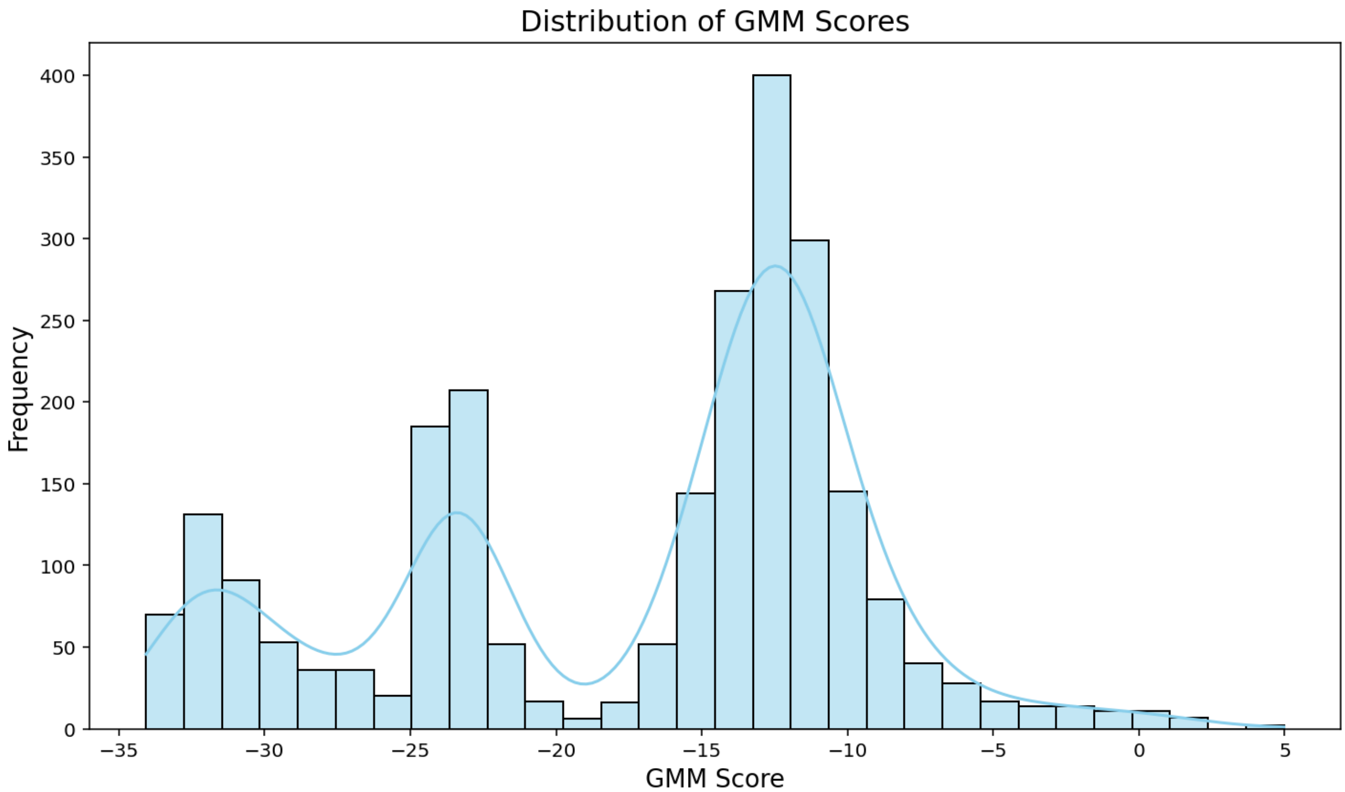

Figure: GMM Score Distribution: Gaussian Mixture Model (GMM) assigns probabilities to data points based on learned distributions. Multiple peaks in the GMM score distribution may reflect subgroup differences, potentially pointing to hidden bias.

4. SHAP-Based Feature Importance

Feature importance analysis using SHAP values identifies which variables contribute most to potential bias.

CRC_incidence_val_Percent - Importance Score: 1.418

Diet low in legumes_Rate_Summary exposure value_val - Importance Score: 0.505

Diet high in sodium_Rate_Summary exposure value_val - Importance Score: 0.493

Low physical activity_Rate_Summary exposure value_val - Importance Score: 0.490

Unsafe water source_Rate_Summary exposure value_val - Importance Score: 0.465

Diet low in calcium_Rate_Summary exposure value_val - Importance Score: 0.424

High body-mass index_Rate_Summary exposure value_val - Importance Score: 0.401

Smoking_Rate_Summary exposure value_val - Importance Score: 0.361

Diet high in processed meat_Rate_Summary exposure value_val - Importance Score: 0.331

Diet low in seafood omega-3 fatty acids_Rate_Summary exposure value_val - Importance Score: 0.306

The figures below show feature importance rankings and SHAP-based explanations.

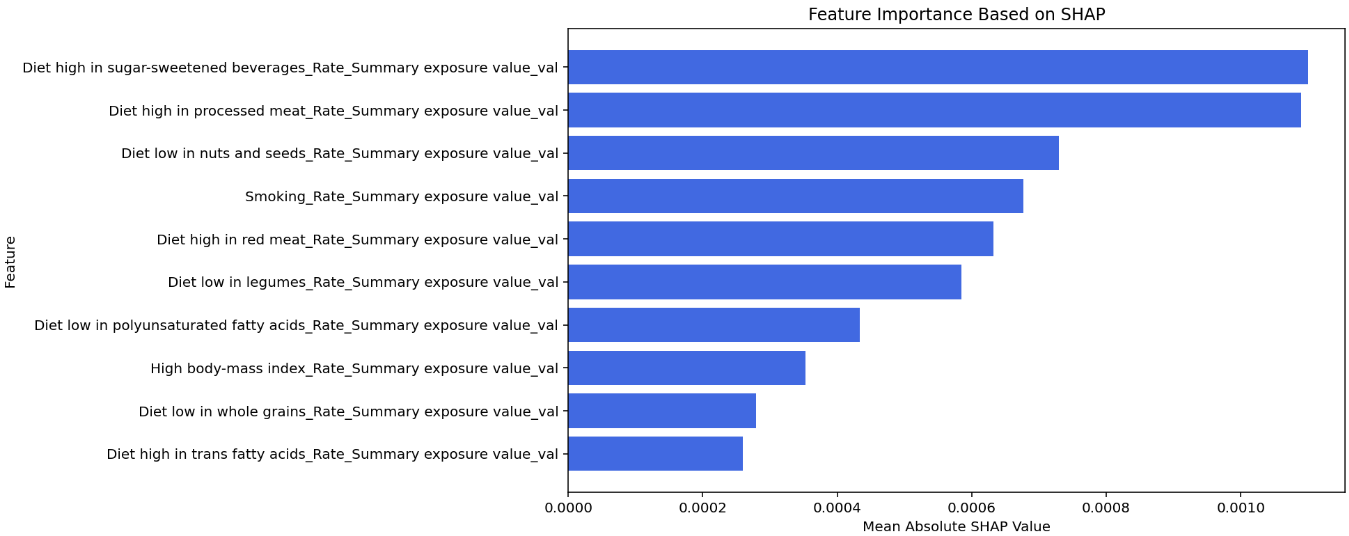

Figure: Shap feature importance. This figure illustrates anomaly scores, helping to identify unusual data points or bias patterns.

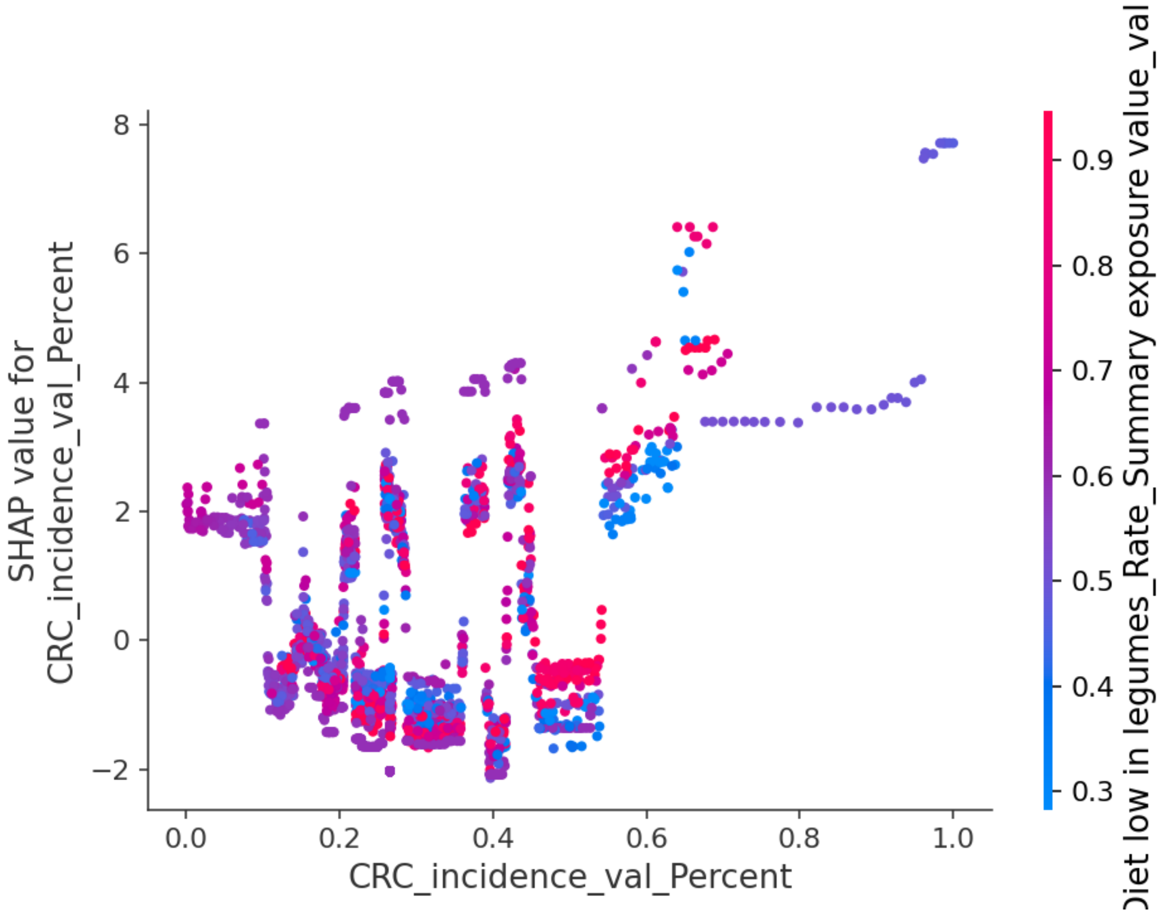

Figure: Shap dependence. This figure illustrates anomaly scores, helping to identify unusual data points or bias patterns.

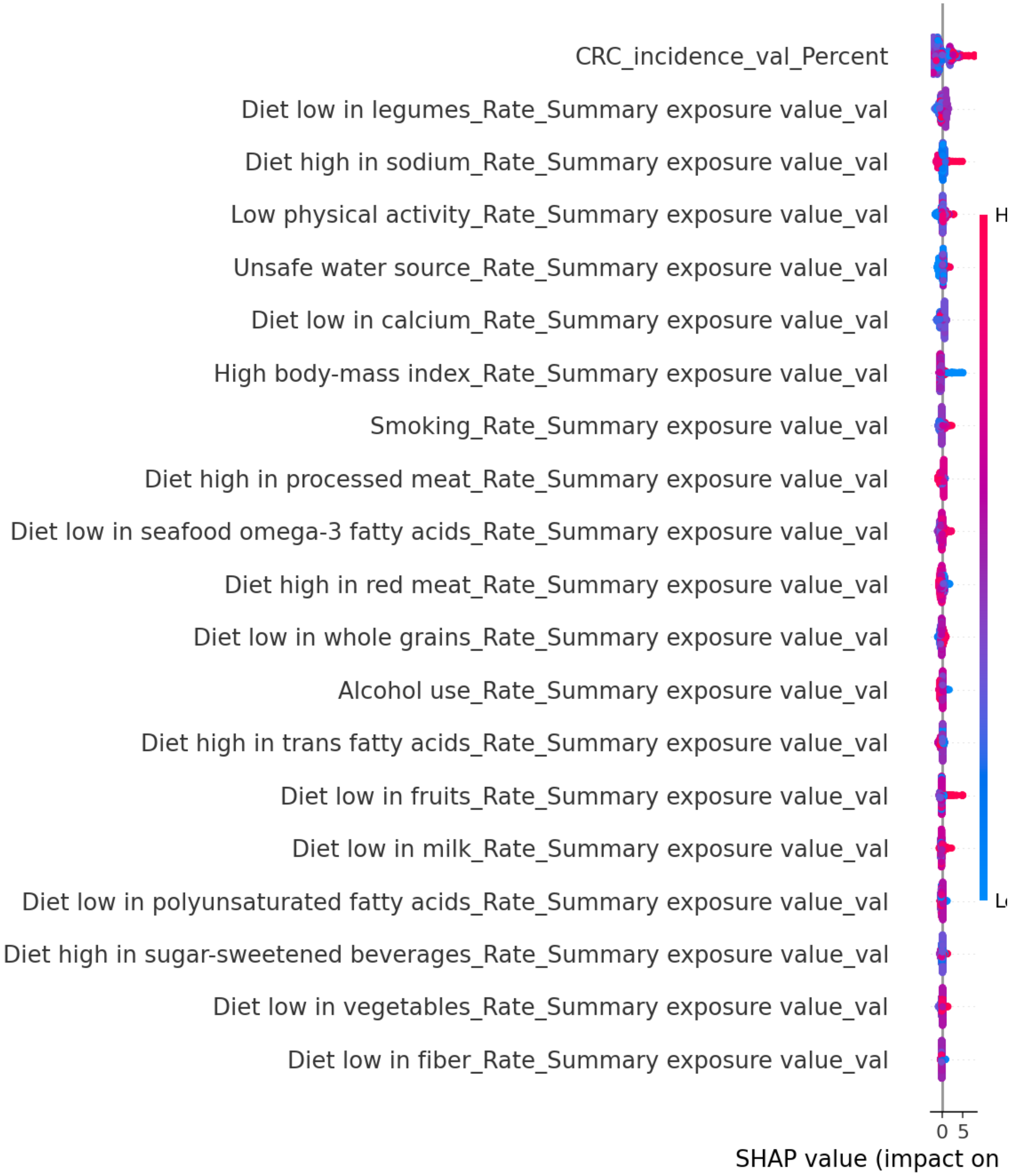

Figure: Shap summary. This figure illustrates anomaly scores, helping to identify unusual data points or bias patterns.

APPENDIX - Explanation of Metrics Used

Kullback-Leibler (KL) Divergence

Measures how one probability distribution diverges from a second reference distribution. High KL indicates significant divergence, signaling potential bias.

Jensen-Shannon (JS) Divergence

A symmetric form of KL divergence that provides a bounded measure (0 to 1) of divergence between distributions. Values closer to 1 indicate significant distributional bias.

Kolmogorov-Smirnov (KS) Statistic

Quantifies the maximum difference between two cumulative distributions, identifying potential distributional biases or shifts.

Total Variation Distance (TVD)

Measures the maximum possible difference between probabilities assigned by two distributions, directly highlighting biased differences.

Lp-Norm

General measure of distance between distributions, where higher values suggest larger divergence or bias in feature distributions.

Autoencoder Reconstruction Error

Quantifies anomalies by measuring how well an autoencoder reconstructs data points. Higher error suggests unusual patterns, possibly indicating bias.

Isolation Forest Score

Assigns anomaly scores based on how easily data points can be isolated; higher scores may indicate unusual distributions or biased patterns.

Gaussian Mixture Model (GMM) Score

Uses probability distributions to model data points. Low likelihood scores highlight unusual data points or potential biases.

SHAP (SHapley Additive exPlanations) Values

Interpret feature importance in model predictions. High SHAP values indicate features significantly contributing to bias or anomalies.